

Data Obfuscation

Sub-techniques (3)



Adversaries may obfuscate command and control traffic to make it more difficult to detect.^[1] Command and control (C2) communications are hidden (but not necessarily encrypted) in an attempt to make the content more difficult to discover or decipher and to make the communication less conspicuous and hide commands from being seen. This encompasses many methods, such as adding junk data to protocol traffic, using steganography, or impersonating legitimate protocols.

ID: T1001

Sub-techniques: [T1001.001](#), [T1001.002](#), [T1001.003](#)

- ① **Tactic:** [Command and Control](#)
- ① **Platforms:** ESXi, Linux, Windows, macOS

Version: 1.2

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[Version Permalink](#)

Procedure Examples

ID	Name	Description
S1111	DarkGate	DarkGate will retrieved encrypted commands from its command and control server for follow-on actions such as cryptocurrency mining. ^[2]
S9003	evilginx2	evilginx2 can modify the Origin and Referrer fields in HTTPS headers it relays between intended victims and legitimate websites to comply with cross-origin resource sharing (CORS) restrictions. ^[3]
S0381	FlawedAmmyy	FlawedAmmyy may obfuscate portions of the initial C2 handshake. ^[4]
S1120	FRAMESTING	FRAMESTING can send and receive zlib compressed data within <code>POST</code> requests. ^[5]
S1044	FunnyDream	FunnyDream can send compressed and obfuscated packets to C2. ^[1]
G0047	Gamaredon Group	Gamaredon Group has used obfuscated VBScripts with randomly generated variable names and concatenated strings. ^[6]
S1100	Ninja	Ninja has the ability to modify headers and URL paths to hide malicious traffic in HTTP requests. ^[7]
S0439	Okrum	Okrum leverages the HTTP protocol for C2 communication, while hiding the actual messages in the Cookie and Set-Cookie headers of the HTTP requests. ^[8]
C0014	Operation Wocao	During Operation Wocao , threat actors encrypted IP addresses used for "Agent" proxy hops with RC4. ^[9]
S0495	RDAT	RDAT has used encoded data within subdomains as AES ciphertext to communicate from the host to the C2. ^[10]
S0610	SideTwist	SideTwist can embed C2 responses in the source code of a fake Flickr webpage. ^[11]
S0533	SLOTHFULMEDIA	SLOTHFULMEDIA has hashed a string containing system information prior to exfiltration via POST requests. ^[12]
S1183	StrelaStealer	StrelaStealer encrypts the payload of HTTP POST communications using the same XOR key used for the malware's DLL payload. ^[13]
S9001	SystemBC	SystemBC has encoded with XOR and encrypted with RC4 its beacon. ^[14]
S0682	TrailBlazer	TrailBlazer can masquerade its C2 traffic as legitimate Google Notifications HTTP requests. ^[15]

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0053	Detect Obfuscated C2 via Network Traffic Analysis	AN0144	Detects excessive outbound traffic to remote host over HTTP(S) from uncommon or previously unseen processes.
		AN0145	Identifies custom or previously unseen userland processes initiating high-volume HTTP connections with low response volume.
		AN0146	Flags unexpected user applications initiating long-lived HTTP(S) sessions with irregular traffic patterns.

References

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